

Review – 1 Document

Design of an Autonomous Floating Embedded System for Pond Hygiene Maintenance

1. Introduction

Autonomous robotic platforms are increasingly being explored for environmental monitoring and maintenance of aquatic ecosystems. Surface-level pollution in open water bodies is primarily driven by floating plastic waste, organic biomass accumulation, suspended particulates, and localized stagnation. Over time, these factors contribute to turbidity increase, oxygen depletion, and microbial growth, resulting in degraded water hygiene.

The objective of this work is to design an autonomous floating embedded system capable of performing continuous preventive maintenance through integrated debris removal, environmental sensing, selective surface-water extraction, and inline filtration. Unlike conventional cleaning mechanisms that rely on manual or episodic intervention, the proposed system emphasizes persistent operation and sensor-driven decision making.

The system is not intended to perform large-scale purification; rather, it aims to maintain baseline hygienic conditions by mitigating contamination at its early stages.

2. Technical Motivation and Regional Context

Temple ponds across Tamil Nadu represent multi-functional water bodies that support irrigation, groundwater recharge, and periodic human interaction. Due to open exposure and frequent organic loading, these ponds experience continuous surface contamination including floating plastic, algae formation, suspended sediments, and poorly circulated regions.

Existing maintenance approaches are largely manual, resource-intensive, and inconsistent. An autonomous embedded platform capable of operating with minimal supervision provides a scalable engineering solution for sustained hygiene management.

3. Review of Existing Work

SMURF: Fully Autonomous Water Surface Cleaning Robot

<https://www.mdpi.com/2077-1312/10/11/1620>

Demonstrates autonomous navigation and coverage path planning for floating debris removal but lacks environmental sensing, extraction mechanisms, and filtration.

Autonomous Water Quality Monitoring and Cleaning for Unmanned Surface Vehicles

<https://www.mdpi.com/1424-8220/21/4/1102>

Integrates sensing and navigation but focuses primarily on monitoring rather than corrective action.

WasteShark – RanMarine Technology

<https://www.ranmarine.io/wasteshark/>

A commercial debris-collection vehicle with strong operational reliability but no onboard contamination mitigation.

OSCAR – Ocean Surface Cleaning Autonomous Robot (VIT Chennai)

https://www.researchgate.net/publication/356514175_Ocean_Surface_Cleaning_Autonomous_Robot_OSCAR_using_Object_Classification_Technique_and_Path_Planning_Algorithm

Validates embedded navigation and waste collection within academic prototypes but does not integrate filtration or decision-based mitigation.

Autonomous Lake Cleaning Robot (India)

<https://www.ijitee.org/wp-content/uploads/papers/v9i4/C8905019320.pdf>

Uses conveyor-based collection with semi-autonomous control but lacks sensor fusion and water-quality intervention.

4. Research Gap

While autonomous navigation, debris removal, and water sensing have individually matured, there is minimal evidence of a unified embedded platform capable of combining debris removal with localized water-quality mitigation through selective extraction and recirculation.

5. System Architecture

The proposed system is a floating embedded platform designed for continuous hygiene maintenance through sensor-driven operation.

Core controller: ESP32-CAM.

The architecture integrates sensors, propulsion motors, pump control, camera telemetry, and a decision engine that activates subsystems based on environmental conditions.

6. Microcontroller Selection – ESP32-CAM

The ESP32-CAM was selected due to its high processing capability, low power consumption, integrated camera, wireless connectivity, and strong embedded suitability.

Compared to Arduino, it provides significantly greater computational headroom for sensor fusion and actuator management. Compared to Raspberry Pi, it maintains far lower power consumption while remaining appropriate for real-time control tasks.

Integrated WiFi enables telemetry, alert generation, and remote diagnostics without requiring additional communication hardware. The controller also supports future upgrades such as lightweight computer vision and adaptive navigation.

The architecture is therefore classified as a General Purpose Processor (GPP)-based embedded system.

7. Operational Subsystems

Navigation: Differential thrust propulsion combined with predefined sweep paths ensures reliable area coverage without requiring computationally expensive SLAM algorithms. Ultrasonic sensors provide obstacle avoidance.

Waste Collection: A wedge-based passive intake directs debris into a mesh storage bin, minimizing mechanical complexity and reducing failure probability. Bin capacity is monitored using load or infrared sensing to trigger maintenance alerts.

Environmental Sensing: Water degradation is detected using proxy indicators rather than chemical analysis. Turbidity serves as the primary parameter due to its strong correlation with suspended particles and organic matter. Temperature sensing assists in identifying stagnant zones.

8. Selective Surface-Water Extraction and Filtration

Surface contamination is typically concentrated within the upper water layer where organic decomposition and oxygen exchange occur. Periodic extraction supports circulation and reduces localized pollutant density.

A low-power submersible pump extracts surface water and passes it through a multi-stage mechanical filter consisting of coarse mesh, foam or sediment filtration, and activated carbon before returning it to the pond.

Returning filtered water prevents payload accumulation and enables continuous operation. This mechanism transitions the robot from passive cleaning toward active environmental mitigation.

9. Feasibility Assessment

The proposed system integrates mature technologies including embedded controllers, turbidity sensors, electric pumps, and floating chassis platforms, resulting in high technical feasibility.

Primary engineering challenges include waterproofing, filter clogging, energy budgeting, and platform stability; however, these are manageable using established embedded design practices.

The system is intended for continuous preventive maintenance rather than rapid purification, ensuring realistic operational expectations.

10. Comparative Advantage

Unlike typical academic prototypes that focus solely on debris removal, the proposed system integrates sensor fusion, decision-based actuation, and inline filtration. The primary contribution lies in system-level integration rather than isolated subsystem development.

11. Limitations

The platform is not designed to purify entire water bodies, remove dissolved chemical pollutants, or perform deep sludge dredging. Instead, it functions as a preventive maintenance system that continuously improves hygienic conditions.

12. Conclusion

This review identified the lack of integrated autonomous platforms capable of combining debris removal with localized water-quality mitigation. The proposed embedded architecture addresses this gap by unifying sensing, actuation, extraction, and filtration within a scalable floating system.

Emphasis is placed on engineering feasibility, modular design, and sustained operation.